Day 137 | Unit 10 | Teacher plan

AASA Applications and Cumulative Review

Connect Equations, Rates, and Percent

Standards: 6.EE.B.6, 6.EE.B.7, 6.RP.A.3b, 6.RP.A.3c

Learning target: I can connect Equations, Rates, and Percent, and explain and check my reasoning.

Cumulative application across enVision Topics 1–8 | Spectrum cross-chapter review

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Define the unknown quantity. Translate the relationship into an equation before calculating. Use the inverse operation and check whether the answer makes sense in the situation. At a constant speed, distance = speed × time. Divide distance by time to find speed, or distance by speed to find time. Keep units consistent before calculating. Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer. Divide the part by the whole, then multiply by 100. Keep the same units in part and whole. The percent can be above 100 if the part is larger than the reference whole.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2, 11, 20

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01 / Learn

Define the unknown quantity. Translate the relationship into an equation before calculating. Use the inverse operation and check whether the answer makes sense in the situation. At a constant speed, distance = speed × time. Divide distance by time to find speed, or distance by speed to find time. Keep units consistent before calculating. Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer. Divide the part by the whole, then multiply by 100. Keep the same units in part and whole. The percent can be above 100 if the part is larger than the reference whole.

Worked example 1

A collection grows by 5 cards to 10 cards. Let x be the starting count. Write and solve an equation.

1. Define the unknown quantity.

2. Translate the relationship into an equation before calculating.

3. Subtract 5 from both sides

4. 5 + 5 = 10.

5. State the result in the form requested: x + 5 = 10; x = 5 cards.

Answer: x + 5 = 10; x = 5 cards

Worked example 2

A quantity changes from a reference whole of 120 to a compared amount of 30. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

1. Divide the part by the whole, then multiply by 100.

2. Percent = (30 ÷ 120) × 100 = 25%.

3. The denominator is the reference whole, not automatically the larger number.

Answer: 25%; denominator 120

Day 137 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A collection grows by 5 cards to 35 cards. Let x be the starting count. Write and solve an equation.

02 / Constructed response

A bus travels at 40 km/h. How long does 200 km take?

03 / Constructed response

50% of a number is 40. Find the number and check.

04 / Constructed response

144 is what percent of 240?

05 / Constructed response

7 identical lengths of rope total 28 m. Let x be the length of one piece. Write and solve an equation, then check.

06 / Constructed response

A bus travels at 40 km/h. How long does 120 km take?

07 / Constructed response

125% of a number is 225. Find the number and check.

08 / Numeric entry and explanation

A quantity changes from a reference whole of 100 to a compared amount of 75. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

09 / Equation entry and two-part response

2 identical ropes have total length 3 meters. Part A: write and solve an equation for one rope length x. Part B: show a substitution check.

10 / Constructed response

A bus travels at 35 km/h. How long does 175/2 km take?

Day 137 | Unit 10 | Answer key

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1. x + 5 = 35; x = 30 cards

Subtract 5 from both sides; 30 + 5 = 35.

6.EE.B.6, 6.EE.B.7 | Constructed response

2. 5 hours

Time = distance ÷ speed = 200 ÷ 40 = 5.

6.RP.A.3b | Constructed response

3. 80

Whole = 40 ÷ 1/2 = 80. Check 50% × 80 = 40.

6.RP.A.3c | Constructed response

4. 60%

144 ÷ 240 × 100 = 60%; the reference whole is 240.

6.RP.A.3c | Constructed response

5. 7x = 28; x = 4 m

Divide the total by 7: 28 ÷ 7 = 4.

6.EE.B.6, 6.EE.B.7 | Constructed response

6. 3 hours

Time = distance ÷ speed = 120 ÷ 40 = 3.

6.RP.A.3b | Constructed response

7. 180

Whole = 225 ÷ 5/4 = 180. Check 125% × 180 = 225.

6.RP.A.3c | Constructed response

8. 75%; denominator 100

Percent = (75 ÷ 100) × 100 = 75%. The denominator is the reference whole, not automatically the larger number.

6.RP.A.3c | Numeric entry and explanation

9. 2x = 3; x = 3/2 meters

Total = number of ropes × one length. Divide both sides by 2; 3 ÷ 2 = 3/2. Check 2 × 3/2 = 3.

6.EE.B.6, 6.EE.B.7 | Equation entry and two-part response

10. 5/2 hours

Time = distance ÷ speed = 175/2 ÷ 35 = 5/2.

6.RP.A.3b | Constructed response